

# Global Trends in CO<sub>2</sub> Emissions, GDP, and Population

## Introduction

Understanding the relationship between environmental impact, economic development, and demographic shifts is vital for addressing global sustainability challenges. This project analyzes comprehensive datasets covering:

- CO<sub>2</sub> emissions per capita
- GDP per capita
- Population
- Continental grouping

The objective is to identify significant patterns and trends through data cleaning, integration, restructuring, and visual exploration. Using available data, this study investigates how various countries and continents compare and change over time.

Central research questions include:

- To what extent is economic growth associated with increased carbon emissions?
- Which nations demonstrate high GDP with relatively low emissions?
- In what ways does population size affect environmental impact?

## Project Structure

This repository contains 8 Jupyter Notebooks, each dedicated to a specific study:

1. **Study\_1\_economic\_growth\_vs\_emissions.ipynb**: Explores the correlation between GDP per capita and CO<sub>2</sub> emissions per capita.
2. **Study\_2\_high\_population\_low\_emissions.ipynb**: Identifies countries with high populations but low per capita CO<sub>2</sub> emissions.
3. **Study\_3\_CO2\_per\_GDP.ipynb**: Analyzes carbon intensity by identifying countries with the highest CO<sub>2</sub> emissions relative to their economic output.
4. **Study\_4\_time\_series\_GDP\_CO2\_population.ipynb**: Visualizes time-series data for GDP, CO<sub>2</sub>, and population for selected countries.
5. **Study\_5\_continental\_emissions.ipynb**: Compares average CO<sub>2</sub> per capita and economic development across continents.
6. **Study\_6\_CO2\_population.ipynb**: Investigates patterns between national population size and total CO<sub>2</sub> emissions using logarithmic scales.
7. **Study\_7\_most\_improved\_countries.ipynb**: Identifies countries that have significantly reduced their per capita CO<sub>2</sub> emissions over time.

8. **Study\_8\_country\_years.ipynb**: CO<sub>2</sub> per capita, and GDP and population for specific country and specific years.

A `modules/data_loader.py` file is used to load and preprocess the data consistently across all studies.

`data` folder contain 4 databases: CO<sub>2</sub> emissions, GDP, population across various countries and a country-continents map.

## Stage 1: Dataset Selection

As mentioned above we used four separate excel sheets, and our final combined data set has more than ten columns and two hundred rows. Datasets were not edited in the excel files, only manipulated using code.

## Stage 2: DataFrame Creation

We used pandas to read the csv files, and the class `DataLoader` to merge the files together. There are multiple merges, such as CO<sub>2</sub> to GDP, result to population, result to continent. After merging the data we called dropped any duplicate rows in the `__main__` using:

```
data.drop_duplicates(inplace=True).
```

The merged data has a hierarchical index of the following:

1. Level 1: Continent
2. Level 2: Country
3. Level 3: year

This was achieved by coding:

```
data.set_index(["Continent","Country","year"], inplace=True)
```

We coded `data.sort_index(inplace=True)` to ensure that the rows are ordered by the hierarchical index.

Finally, we used `pd.to_numeric()` function and `dropna(subset=[...])` on each data frame before merging to ensure that any missing data or non numeric values are converted to NaN then dropped.

## Stage 3: User Entry

The user is required to navigate a menu and a sub-menu through the terminal by entering either numbers or letters, if the user enters an invalid number or character they are prompted to try again. With the data set the user is prompted multiple times to enter either a combination of year(s), country(ies), or both. In this step as well, the user will be prompted to enter a valid entry.

## Stage 4: Analysis and Calculations

After cleaning and merging the .csv files the describe method is called in the main and it is printed in the console

```
print(data.describe())
```

Head is also printed for 5 rows

```
print(data.head(5))
```

In the data\_loader.py the method load\_and\_prepare\_data() we compute total\_co2 and total\_gdp by multiplying the co2 and gdp with population.

```
data["total_co2"] = data["co2"] * data["population"]
```

```
data["total_gdp"] = data["gdp"] * data["population"]
```

This gives us two new columns in the merged data frame.

In study 1, we used aggregation and groupby to determine the correlation trend.

```
data.groupby("year").apply(lambda g: g["gdp"].corr(g["co2"]))
```

This aggregates the data per year subsets.

We used masking quite a bit in our program, however, one of the examples is the following.

```
Data[data["year"] == year]
```

In the main, we created the pivot table and printed it to the user.

```
pivot_table_co2 = data.pivot_table(values='co2', index='Continent', columns='year',  
                                   aggfunc='mean')
```

```
print(pivot_table_co2)
```

Our dataLoader class has two methods called def conversion(...) that modifies the population into numbers, and def load\_and\_prepare\_data(...) that cleans and merges the .csv files.

## Stage 5: Export and Matplotlib

At the end of our program we exported the data set into an excel file.

```
data.to_excel('cleaned_merged_dataframe.xlsx', index=True)
```

This ensures that the hierarchical index and header values are maintained.



### Data Sources:

- CO<sub>2</sub> Emissions  
Source: Gapminder, "Download the Data," Gapminder.org, 2019.  
<https://www.gapminder.org/data/>

- GDP per Capita  
Source: Gapminder, "Download the Data," Gapminder.org, 2019. <https://www.gapminder.org/data/>
- Population  
Source: Gapminder, "Download the Data," Gapminder.org, 2019. <https://www.gapminder.org/data/>
- Country–Continent Mapping  
Source: Kaggle, "Datasets | Kaggle," Kaggle.com, 2024. <https://www.kaggle.com/datasets>